

RTIPS

8/1/24

I/P $\rightarrow$ Process $\rightarrow$ O/P

$\downarrow$
some transformation inside
is a transfer function

} Process

System $\rightarrow$ Hardware $\rightarrow$ sensors eg thermocouple
 $\downarrow$ + software
required to monitor, control, optimise a process
more about software

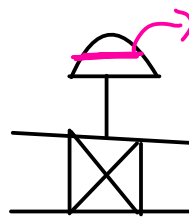
system $\rightarrow$ a combination
of hardware, software
for monitoring, controlling
optimising a process

Two way highway $\rightarrow$ signal from process to CR
" " CR " process
sensor - transmitter - control room

} to maintain
highway you need
hardware

pneumatic $\rightarrow$ 3-15 psi
- 4-20 mA

air to open valve
air to close valve



diaphragm

air $\downarrow$
air close

air $\uparrow$
air open

Data Acquisition system $\rightarrow$ software part

$\hookrightarrow$ SCADA ?

Intelligent part - optimisation, fault detection, image detection
of the system

$\hookrightarrow$ process will remain same (as it is), we control the system

What is Real Time?

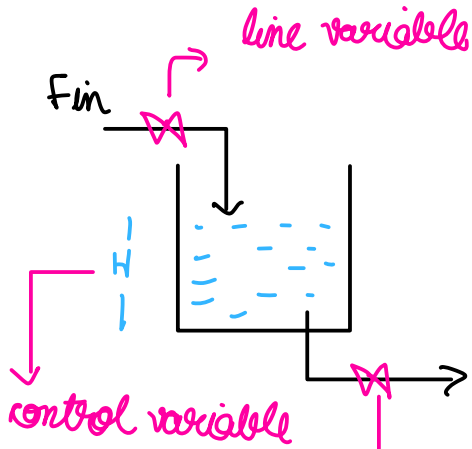
Real, but not live used for training, backtesting

→ cricket example → YT video → old (off-line)

TV → one way communication (on-line)

live but can't interfere ← On ground

→ two way communication (Real time)



Mass in - Mass out

= accumulation

If we put sensors → level sensor etc. (control variable, bad & manipulating variable)

the process is the same

Suppose we plot F_{in} , F_{out} , h vs time

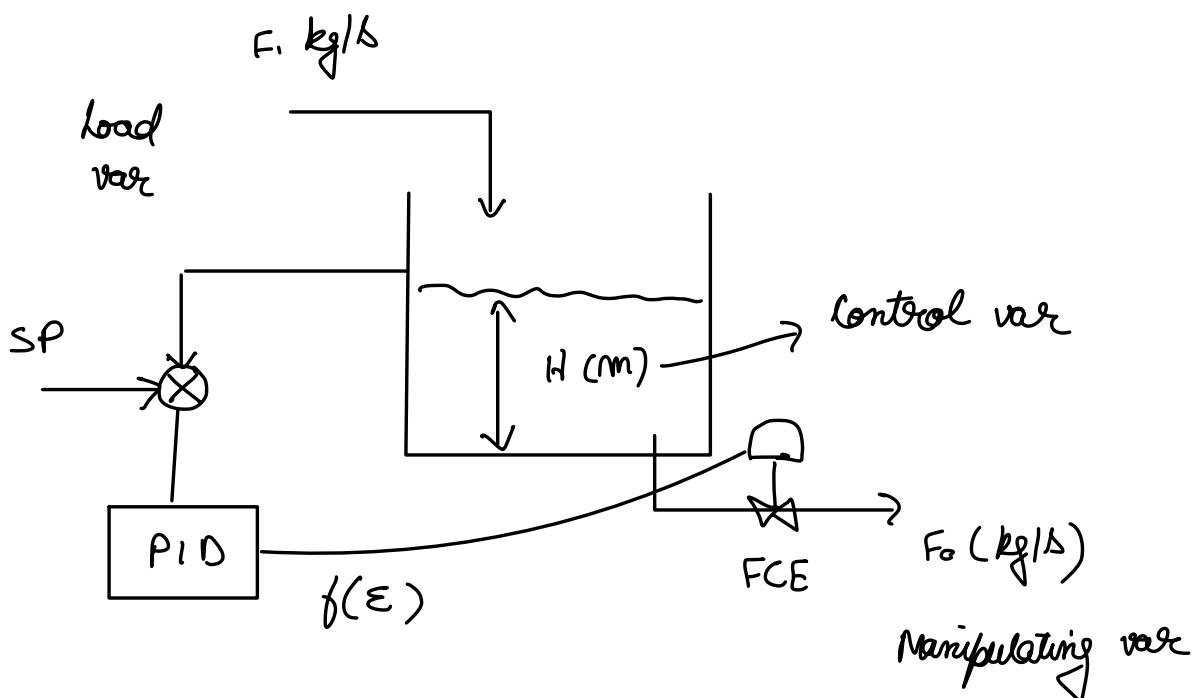
If we use model and run, we can generate 10 mins data maybe in 5 seconds

→ this is offline

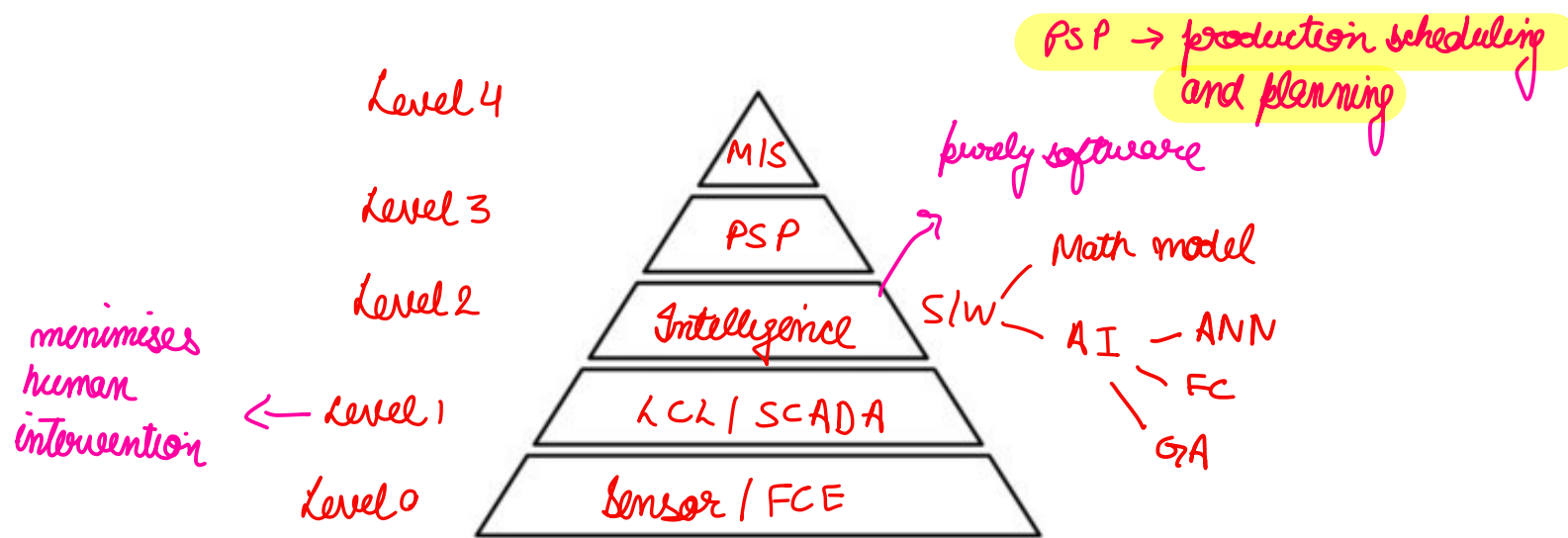
Whereas in online → 10 mins data take 10 mins lagence

Real time → same clock in both parts

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Automation hierarchy

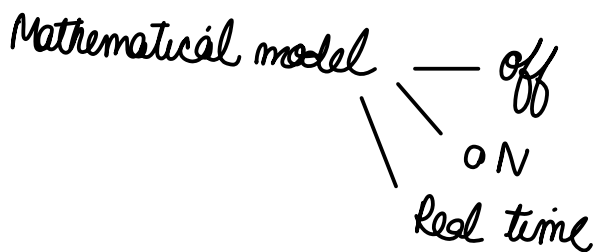


Level 2:

Time	F_1	F_0	H
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-

→ A change in F_1 , will not be immediately reflected in F_0 , there will be a time lag. So our time stamps will have to be adjusted according like

ex In furnace, change in amount of raw material might be reflected 6-8 hours after in products / outstream



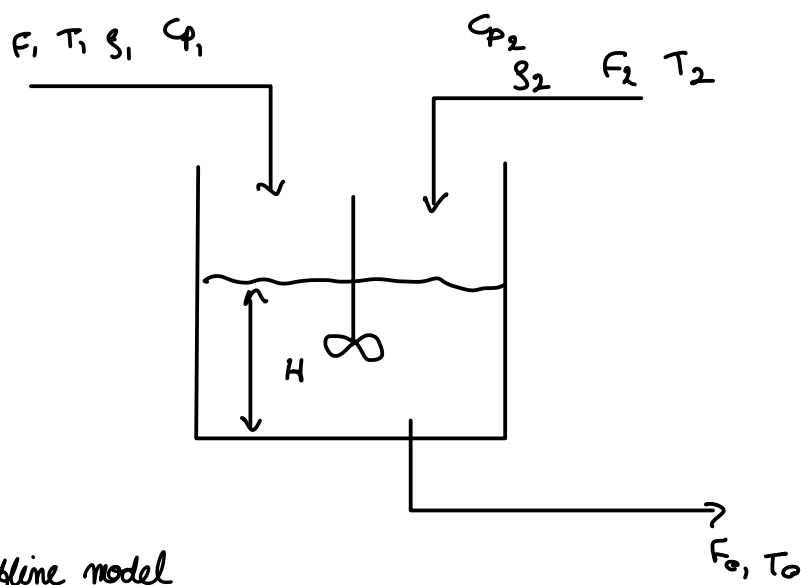
Model → maths, equations related to a process

↳ can't be perfect (assumptions are there)

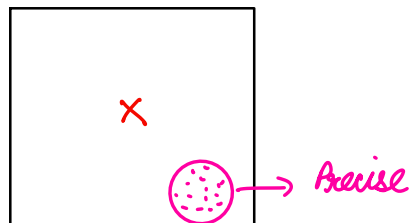
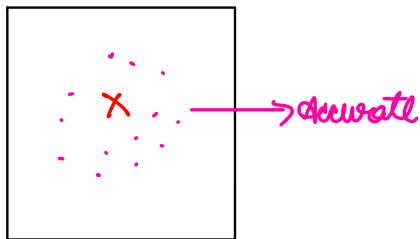
→ ODE, PDE, Algebraic

Model = Equations + Assumptions + BC + IC

↳ To solve the model, we need an algorithm → then simulation



1. You make offline model
2. Then check with past data from the plant

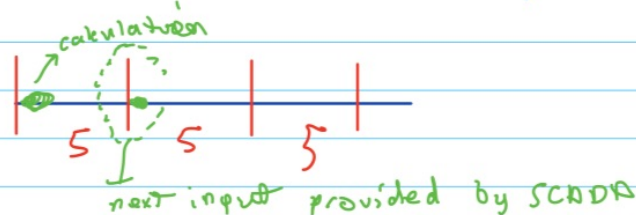


Accuracy

Precision

Offline: Back testing
/ R.T.

Online: Scada provides data to the models from the process, you calculate the output and then measure it with the actual output.



Model should predict H_{next} after 5 seconds, and it can then be used as the S.P. for level 1.

$$F_i - F_o = \frac{dH}{dt} (CpP)$$

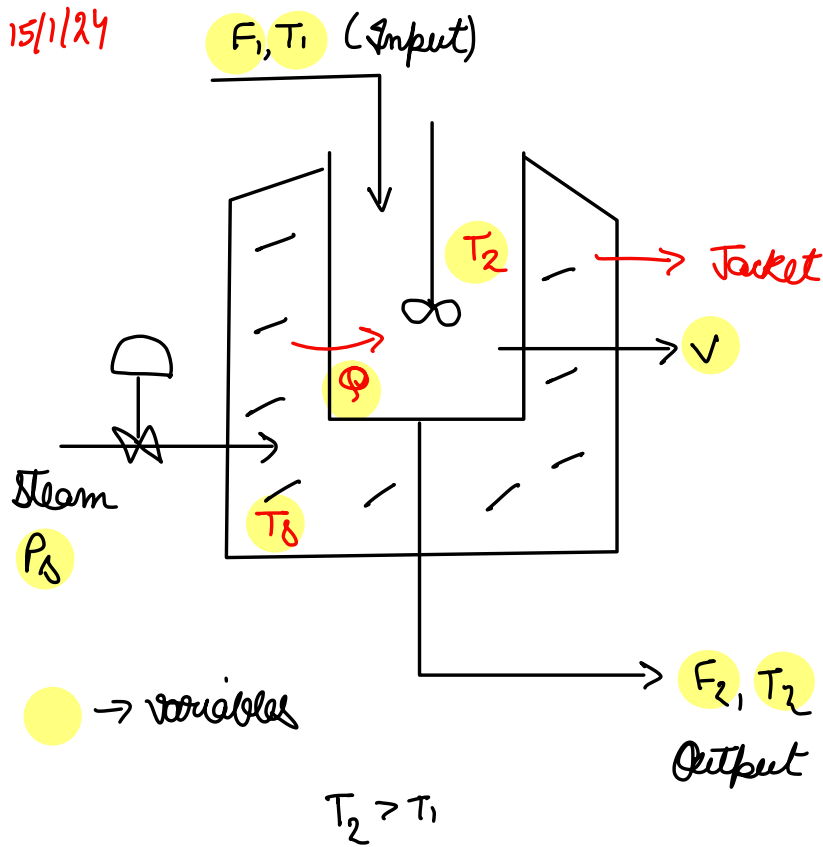
$$\frac{dH}{dt} = \frac{H^{n+1} - H^n}{\Delta t}$$

Now back-calculate to control F_o .

This model is a CONTROL MODEL.

A SIMULATION MODEL will actually calculate H .

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when you want to use steam for heat transfer

→ use saturated steam

when you want to use steam for work → use superheated steam (ex Thermal power plant)

Why? To avoid condensation (can damage turbine)

In this jacket system we want to use the huge amount of latent heat, so we want condensation to happen inside jacket, so that huge amount of heat is released.

Steps for modelling

1. Identify the variables
2. Take the assumptions
3. Equations

(i) Mass balance : $F_1 - F_2 = \frac{dV}{dt}$

(ii) Heat balance : $F_1 C_p T_1 - F_2 C_p T_2 + \frac{Q}{s} = \frac{d}{dt} (V C_p T_2)$

(iii) Jacket $T_s = f(P_s)$ [from Steam Table]

(iv) Jacket HT = $U A (T_s - T_2)$

↓
overall heat transfer coefficient

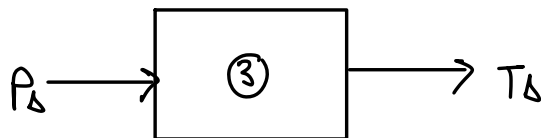
Information Flow Diagram

Each equation is represented as a block, with input, output and connections

Thumb Rule

- Whenever there is a time derivative, ODE, it will be output

$\frac{d}{dt} \rightarrow$ will be output



$$F_1^n - F_2^n = \frac{V^{n+1} - V^n}{\Delta t}$$

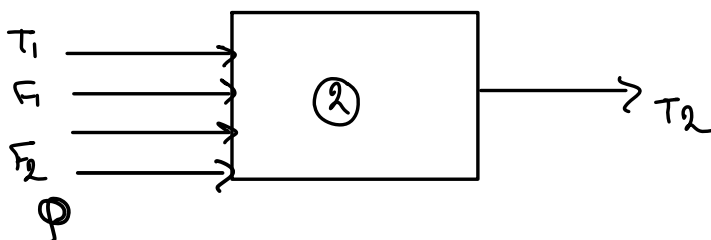
$$V^{n+1} = V^n + (F_1^n - F_2^n) \Delta t \quad (\text{explicit scheme}) \rightarrow \text{can be written explicitly}$$

$\hookrightarrow$ offline calculation

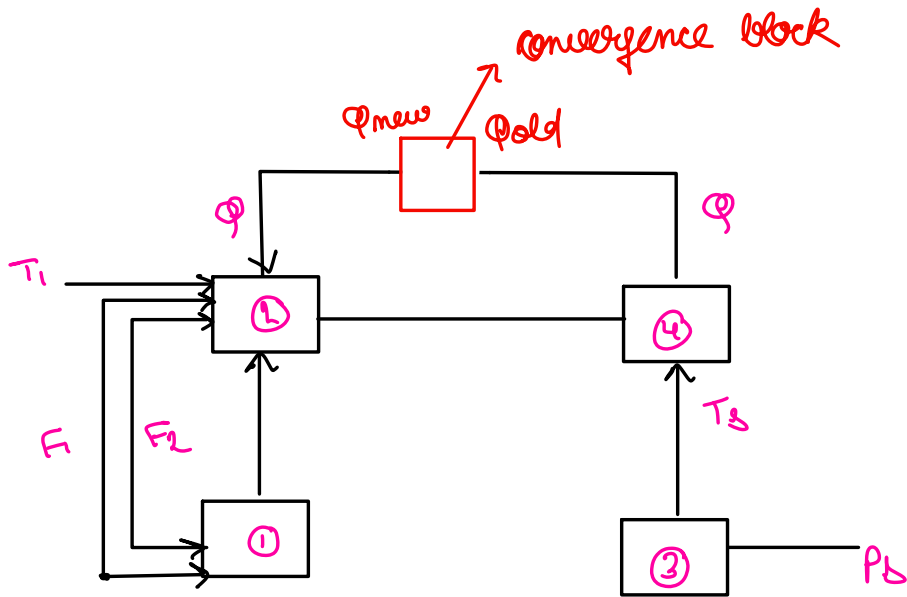
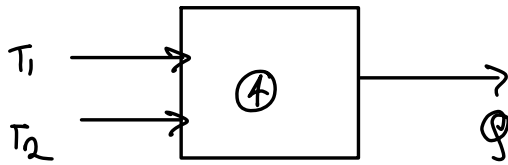
(Predicting)

$$\underbrace{F_1^{n+1} - F_2^{n+1}}_{\text{coming from plant}} = \frac{V^{n+1} - V^n}{\Delta t} \rightarrow \text{definition of forward difference won't change}$$

For implicit scheme $\rightarrow$ matrix inversion is involved



Did not take V , as V already got covered in output of 1, so only T_2



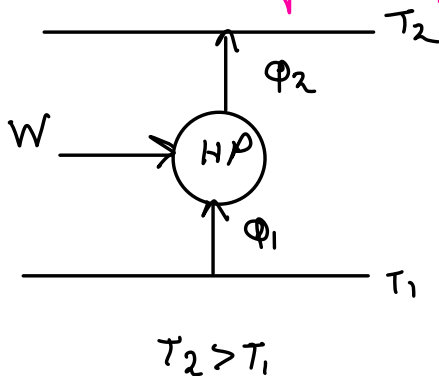
convergence block $\rightarrow$ to tear a variable into two and break a loop

Tear variable $\rightarrow Q$

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Heat

- Work is input $\leftarrow$ Pump
- Heat is taken from lower temp to higher temp

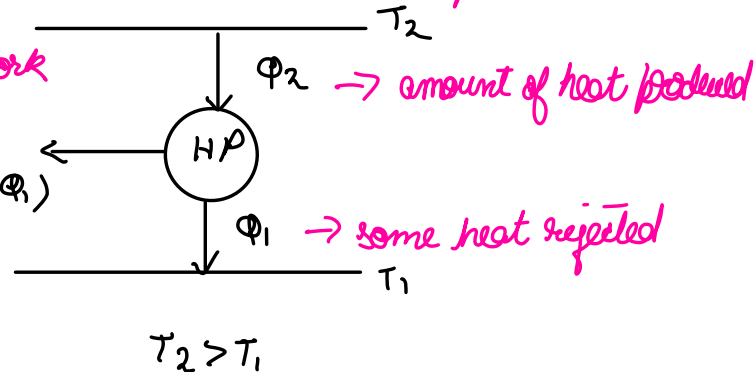


Engine $\rightarrow$

- Work is Output
- Heat is taken from higher temp to lower temp

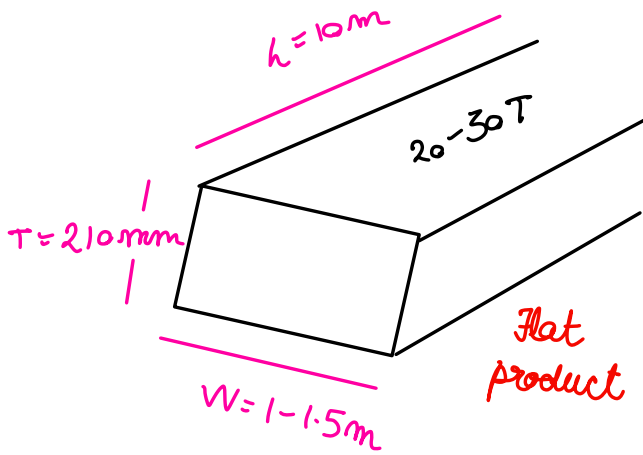
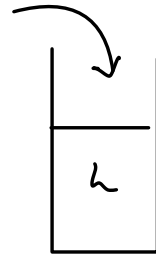
remaining taken as work

$$W = (Q_2 - Q_1)$$

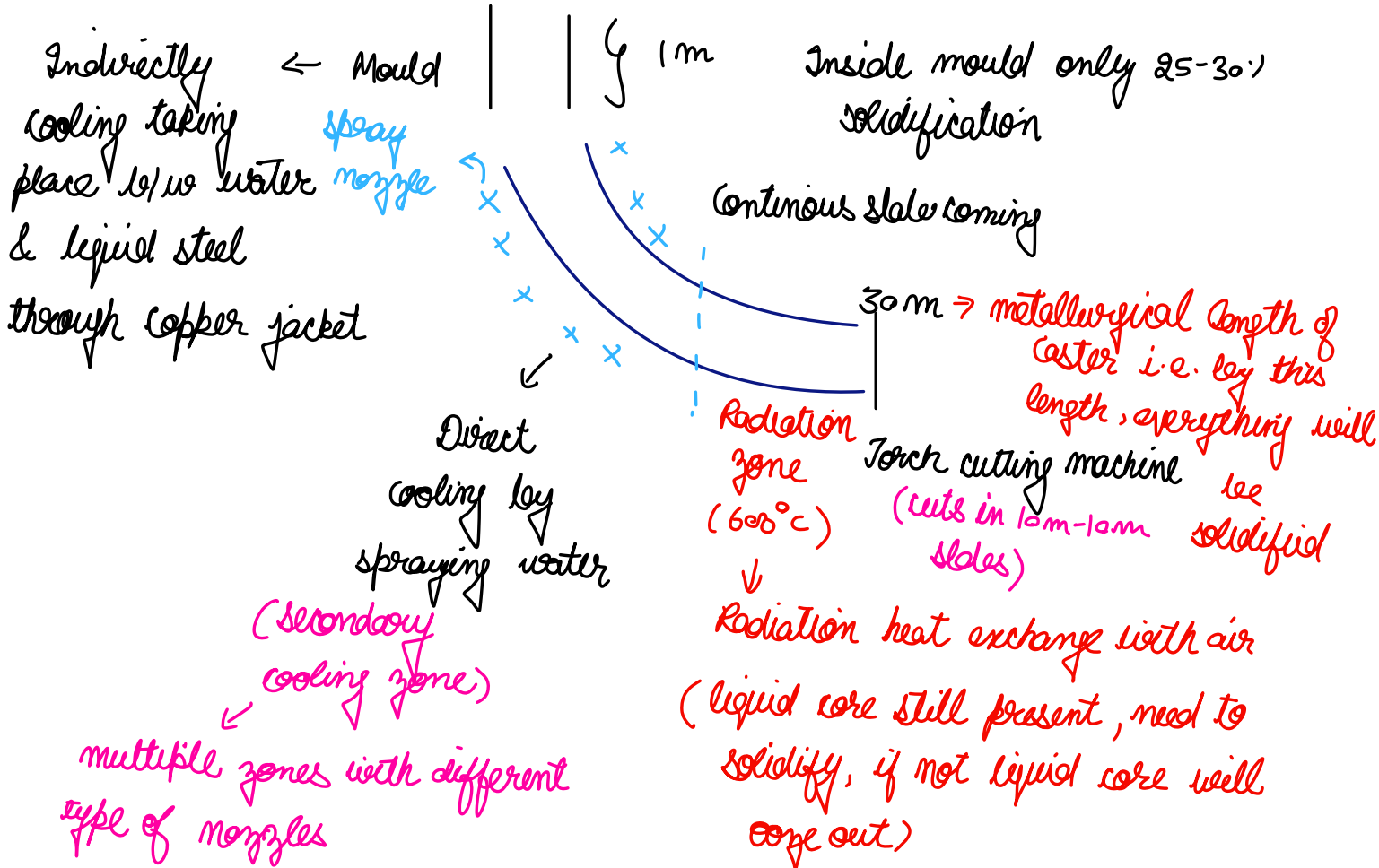
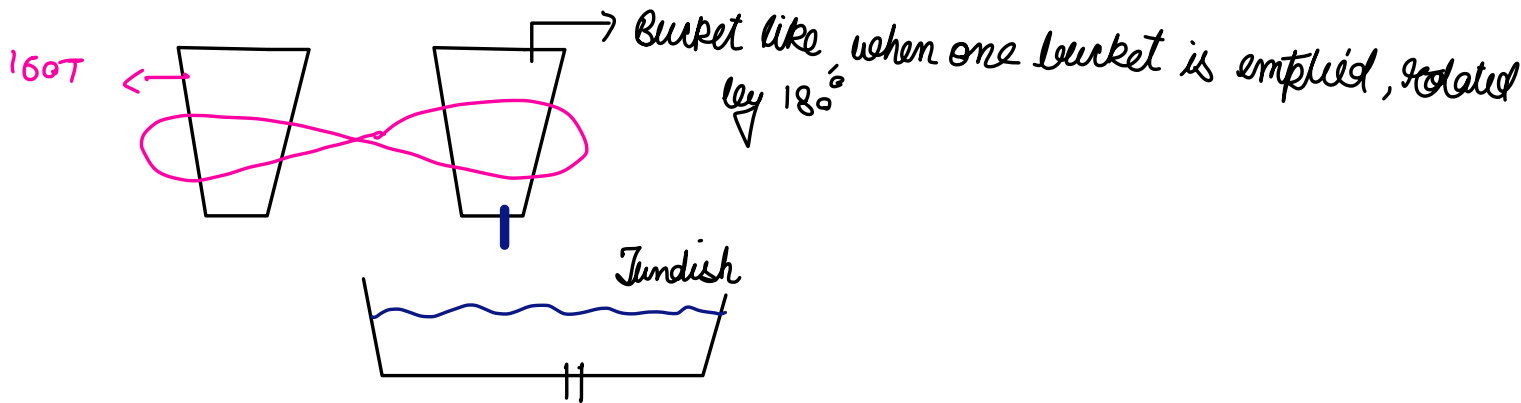


Slab Caster

- Continuous slab caster
- Liquid metal coming in, slab going out



Each casting machine produces 2 mTPA



$V_{\text{cast}} = \text{casting speed} \approx 1-1.5 \text{ m/min}$

If $V_{\text{cast}} \uparrow$, shell thickness $\downarrow$, chances of breakout $\uparrow$

If $V_{\text{cast}} \uparrow$, metallurgical length $\uparrow$

coefficient

By manipulating water flow rates, we try to control effective heat transfer $\uparrow$ and hence the temperature of our steel and its solidification

conduction $\rightarrow$ molecular

convection $\rightarrow$ bulk motion

$\hookrightarrow$ here due to existence of V_{cast}

$$\underbrace{k \left(\frac{\partial^2 T}{\partial x^2} \right) + k \left(\frac{\partial^2 T}{\partial z^2} \right)}_{\text{conduction}} = \underbrace{\rho C \frac{\partial T}{\partial t}}_{\substack{\downarrow \\ \text{temporal} \\ \text{term}}} + \underbrace{\rho V_{\text{cast}} C \frac{\partial T}{\partial x}}_{\substack{\text{mass flow rate} \\ \text{convective part}}}$$

Liquidus temperature $\rightarrow$ temp when first drop solidifies

Solidus " $\rightarrow$ temp when whole slab solidifies

We have taken each segment as 0.5 m (segment tracking)

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$$\underbrace{\rho C_p \frac{\partial T}{\partial t}}_{\text{Transient}} + \underbrace{\rho V \frac{\partial T}{\partial t}}_{\text{advection}} = \underbrace{k \left(\frac{\partial^2 T}{\partial x^2} \right) + k \left(\frac{\partial^2 T}{\partial y^2} \right)}_{\text{conduction}}$$

heat needed to be extracted to cool steel

$$\dot{m} (C_p \Delta T + \lambda) \approx 40 \text{ MW}$$

↗ latent

↘ sensible

$$V_{\text{cast}} \times \text{area} \times \text{density} = \text{mass flow} \quad (\dot{m})$$

$$1.5 \text{ m/min} \mid 210 \text{ mm} \times 1.5 \text{ m} \mid 7400 \text{ kg/m}^3 = 58.275 \text{ kg/s}$$

$$\lambda = 272 \text{ kJ/kg}$$

$$T_{\text{start}} = 1150^\circ\text{C} \text{ (liq)} \quad T_{\text{end}} = 600^\circ\text{C} \text{ (sol)}$$

$$C_p = 750 \text{ J/kg K}$$

$$\dot{m} (\lambda + C_p \Delta T)$$

$$= 39906.35 \text{ kJ/s} \quad 39.90625 \text{ MW}$$

10% in mould region
60% in secondary cooling zone
30% atmosphere

}

% of heat extracted in different regions

- as residence time ↑, cooling efficiency ↑
- as dimension of slab ↑, V_{cast} has to be reduced, to ensure proper cooling of slab

Water used to remove the heat

→ control variable

Boundary condition — Dirichlet — Temperature
 — Neuman — Temperature gradient [heat flux]

T → Temperature profile
CV (Control Variable)

$$q = A - b\sqrt{t}$$

→ const → residence time
→ for mould
→ heat extracted by mould

Manipulated Variable → water flow rate

Simulation & Control Mode

↓
Temperature profile is the output

↘ Input Temperature profile

— Empirical relation of water spray

$$q = A(\dot{w})^m$$

↘ water flow rate
↘ lit/min / m³ in

A, m
↘ provided by manufacturer

FTCS → forward time step
control space

ADI (Alternate Direct Implicit)

$$1D \rightarrow \frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} \right)$$

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \left(\frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2} \right) \rightarrow \text{stability analysis has to be done}$$

(explicit)

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \left(\frac{T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}}{\Delta x^2} \right)$$

(implicit)

— explicit → stability issue

implicit → intrinsically stable

instability issues can be handled using small time steps

Implicit always stable $\rightarrow$ system of equations

Tridiagonal matrix formed $\rightarrow$ solved using thomas algorithm

$$b T_{i-1}^{n+1} + d T_i^{n+1} + a T_{i+1}^{n+1} = c$$

$\hookrightarrow$ includes T_i^n (known from previous iterations)

$$d_j = d_j - \frac{b_j}{d_{j-1}} a_{j-1}$$

$$a_j = c_j - \frac{b_j}{d_{j-1}} c_{j-1}$$

} for loop

$\nearrow$ above diagonal

$$a = b = -\alpha \frac{\Delta t}{\Delta x^2}$$

$\downarrow$

below diagonal

$$u_{nj} = c_{nj} / d_{nj}$$

$$u_k = \frac{c_k - a_k u_{k+1}}{d_k}$$

$$d = \left(1 + 2\alpha \frac{\Delta t}{\Delta x^2} \right)$$

$\downarrow$
diagonal

2D domain

TDMA $\rightarrow$ 1D

there will be 5 unknowns at any point

$\downarrow$

$j+1, i, j+1, i+1, i-1$

$x \rightarrow$ implicit direction

$y \rightarrow$ explicit direction

(sweeping) $\longrightarrow$

$\downarrow$
1 cycle

$\hookrightarrow$ 2 vertical
2 horizontal

$\downarrow$ (obtained from previous equations)

5 variables reduced to 3 variables

$\downarrow$

application of thomas algorithm

$B \rightarrow T$
 $R \rightarrow L$
 $T \rightarrow B$
 $L \rightarrow R$ } cycle

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Fuzzy logic
└─
Crisp

→ Fuzzy logic with engineering application by Timothy J Ross

$A \rightarrow \{1, 2, 5, 10\} \rightarrow 100\% \text{ member of } A$

Fuzzy A $\rightarrow \{1, 2, 5, 10\} \rightarrow \text{not } 100\% \text{ member, partial members}$
 $\quad \quad \quad \underline{0.2} \quad \underline{0.4} \quad \underline{0.8} \quad \underline{0.1}$

└─ partial membership (like dual citizenship)

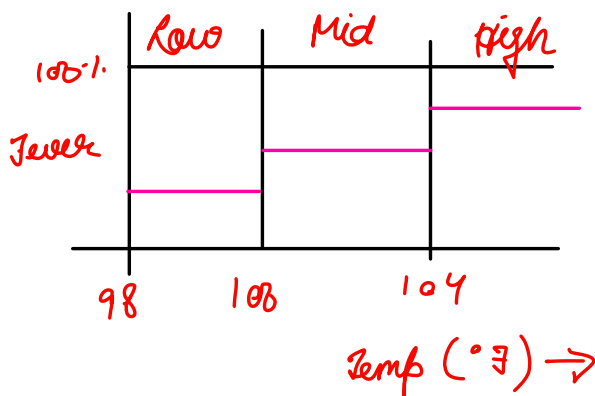
One member maybe member of different sets with partial membership

Membership function $\rightarrow$

Linguistic Variable $\rightarrow$ classification of a variable, without any number but in language

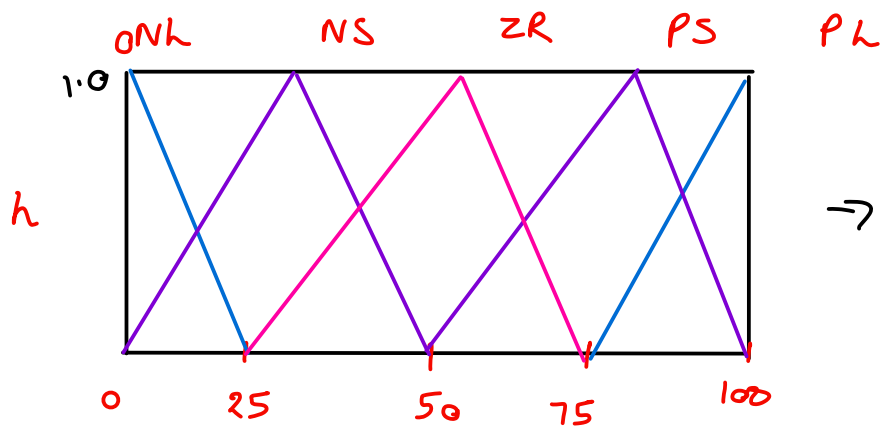
ex fever $\begin{cases} \text{low} \\ \text{moderate} \\ \text{high} \end{cases}$

Then what is low, moderate, high? $\rightarrow$ Membership function governs it



$\rightarrow$ not well defined
vertical lines,
crisp logic,
crisp lines

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→ this is overlap of all 5 functions

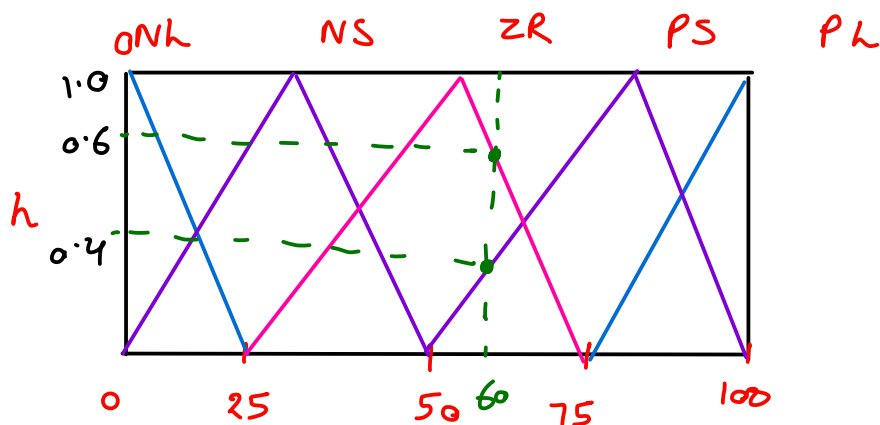
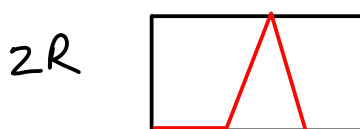
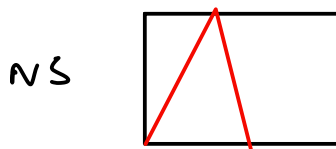
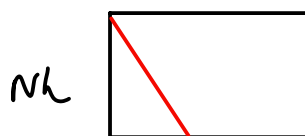
ONK → -ve large

NS → -ve small

ZR → zero

PL → +ve large

PS → +ve small



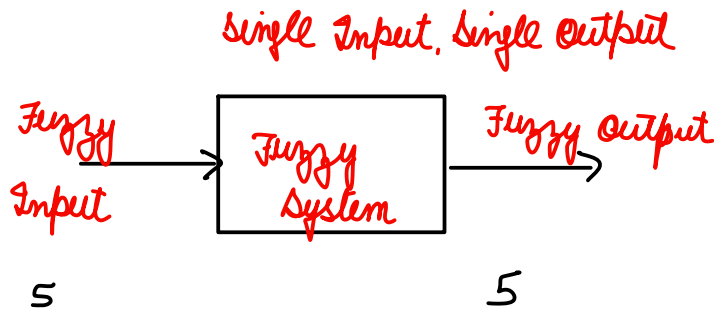
Let crisp input = 60%.

Draw vertical line → Interactions with only two classes

Rest classes membership value is zero

Q Find fuzzy output for a crisp input (For exams)

Fuzzy Rulebook



For 5 FI, 5 FO, 25 combinations possible

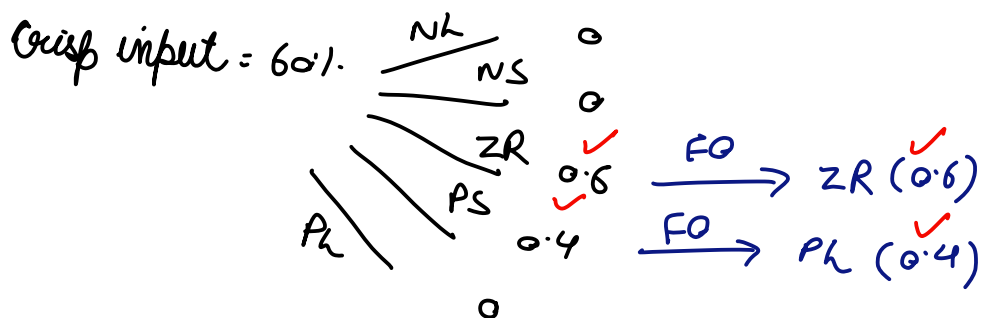
Rulebook $\rightarrow$ in my own hands, a tuning parameter

Rule	IF (FI)	THEN (FO)
1	NL	NL
2	NS	ZR
3	ZR	ZR
4	PS	PL
5	PL	PL

Out of 25 available options, the designer selected 5, depends on the designer

$\rightarrow$ Fuzzy output is also a class with a membership value

Only a non zero membership value will trigger the rulebook

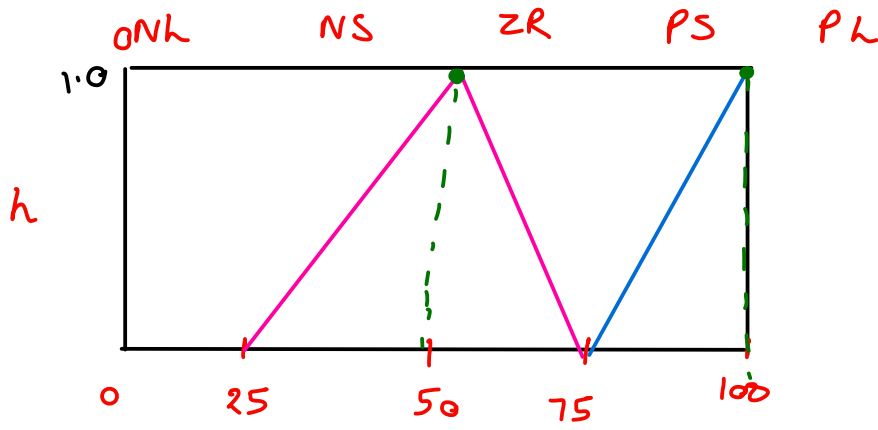


For SISO, membership value of input = membership value of output

If system is a two input one output $\rightarrow$ then lower value of membership will be allotted to output

Chain weakest $\leftarrow$
as weakest link rule example

De-Fuzzification



→ Removed 0 membership
whole classes

• → highest points

1. Method of height
2. Method of centre of gravity (more robust)

F0 ZR (0.6)
 PL (0.4)

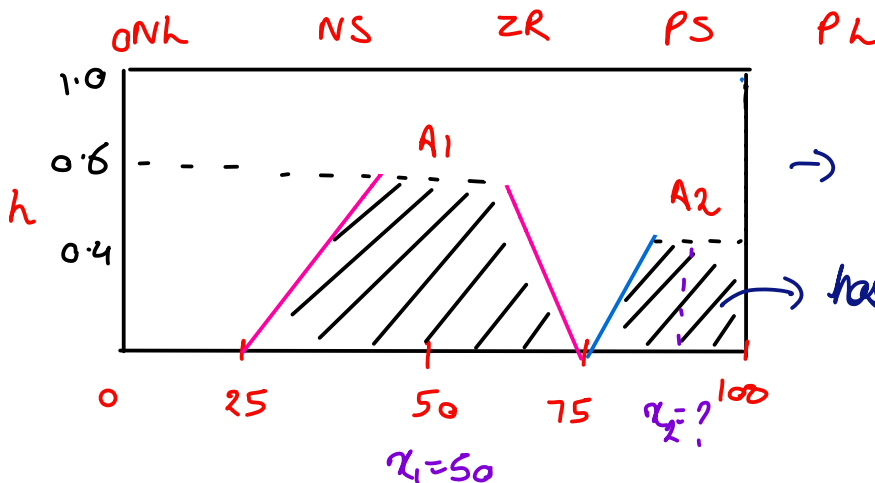
Number → Class (Fuzzification)
Class → Number (De-fuzzification)

1. Find highest point

$$\frac{\sum \mu_i h_i}{\sum \mu_i}$$

$$\begin{aligned} \text{F0 } \text{ZR (0.6)} &\rightarrow 50 \times 0.6 \\ \text{PL (0.4)} &\rightarrow 100 \times 0.4 \end{aligned} \quad \begin{array}{l} \nearrow \\ \searrow \end{array} \sum \rightarrow 70$$

In method of height → shape of the curve does not take in affect,
simply peaks used, so you can't tune anything



→ chopped of top portion

→ hashed / shaded area

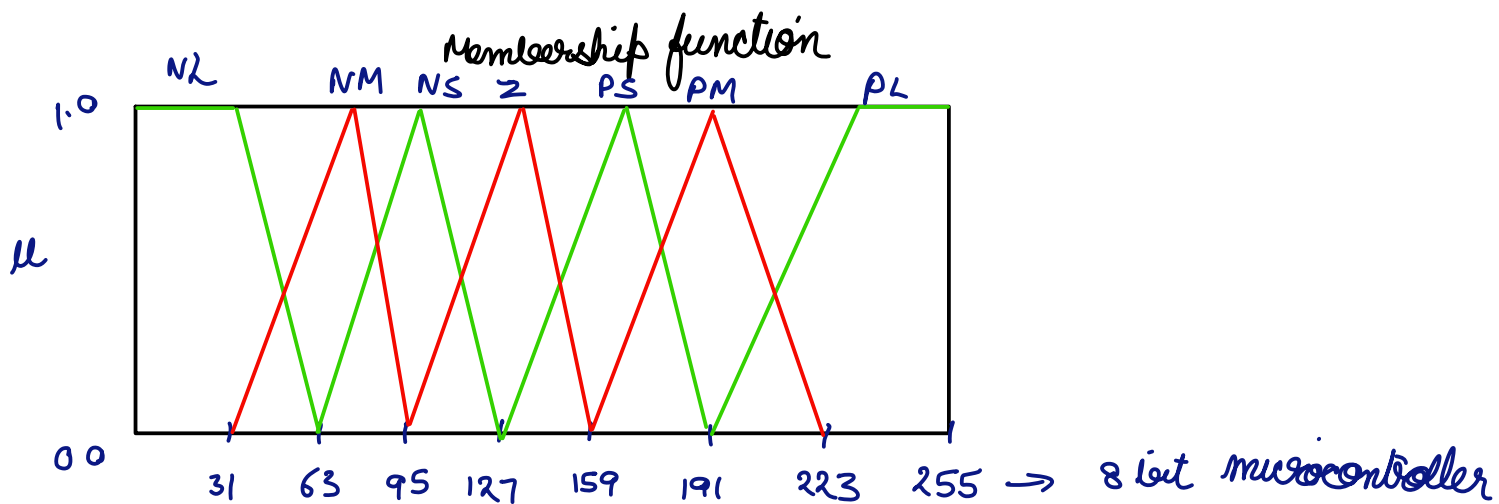
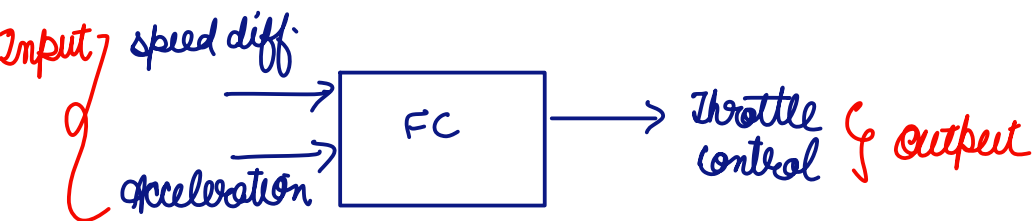
$x_1 = 50$ (location of CG of that area, here since A_1 is symmetric, $x_1 = 50$)

$$CG \Rightarrow x^* = \frac{\sum A_i x_i}{\sum A_i} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2}$$

$\downarrow$
 CG of the system

7/2/24 extra class

Fuzzy controller



symmetric membership fn $\rightarrow$ at any pt you draw vertical line, sum will always be 1

$\hookrightarrow \sum \mu = 1$

N $\rightarrow$ -ve

P $\rightarrow$ +ve

L, M, S $\rightarrow$ large, medium, small

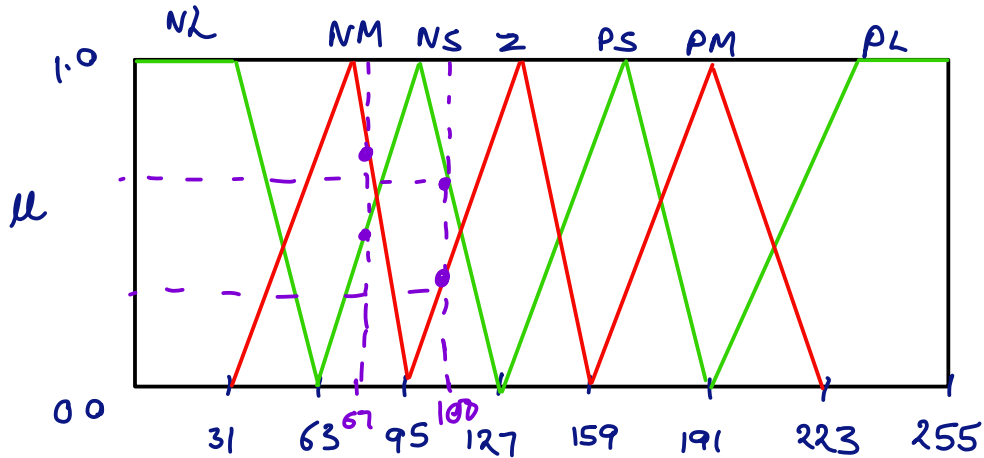
NL, PL $\rightarrow$ Trapezium type

Rest $\rightarrow$ Triangular

Input

speed diff = 100

acc = 67



Two membership classes $\rightarrow$ Z, NS

speed diff = 100 Z (0.1563) $\rightarrow$ get one

NS (0.8437) $\rightarrow$ other value get by subtracting from 1

acceleration = 67 $\rightarrow$ NS (0.125)
NM (0.875)

[Fuzzification of a crisp number]

Rule book

Rule # if (CSD) and (ACC) then (THROTTLE)

Theoretically 7^3 rules possible $\rightarrow$ 7 speed difference
7 acceleration
7 throttle } 3 type

Rule #	If $\mu(SD)$ and $\mu(ACC)$ then Throttle		
1	NL	Z	PL
2	Z	NL	PL
3	NM	Z	PM
4	NS	PS	PS
5	PS	NS	NS
6	PL	Z	NL
7	Z (0.1563)	NS (0.125)	PS
8	Z (0.1563)	NM (0.875)	PM

$7 \leftarrow Z-NS \rightarrow PS$ (0.125)
 $8 \leftarrow Z-NM \rightarrow PM$ (0.1563)

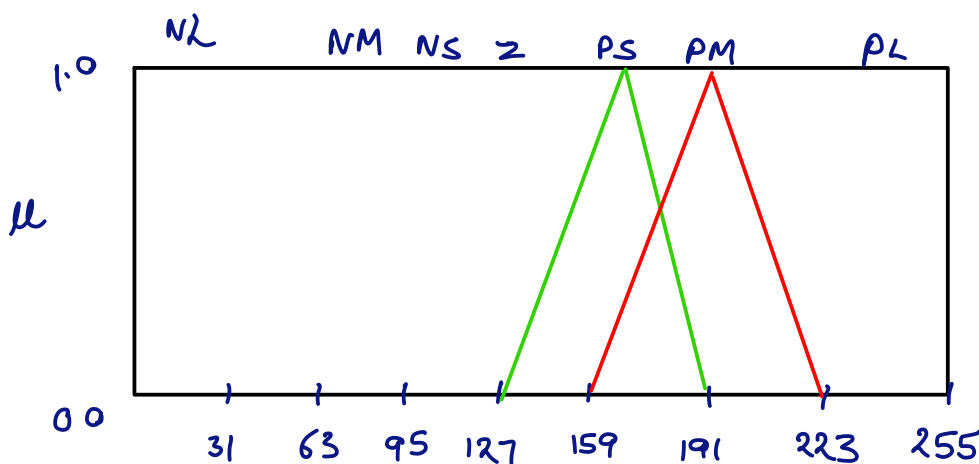
use weakest link rule to assign value [Min. of the input]

NS-NS
 NS-NM

Now have to do de-fuzzification to get crisp output

In output $\rightarrow$ if there are more than one same class is appearing you take the maximum

In Input $\rightarrow$ you take minimum
 (MIN-MAX Method)



$$\frac{\sum \mu_i h_i}{\sum \mu_i}$$

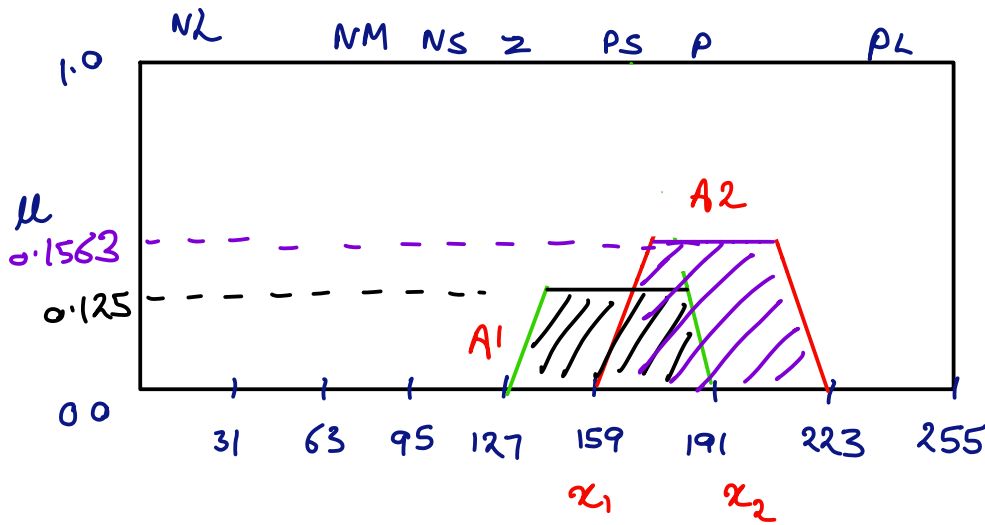
By method of height

$$\begin{aligned}
 159 \times 0.125 &= \\
 191 \times 0.1563 &=
 \end{aligned}$$

$$\left\{ \frac{\sum \mu_i h_i}{\sum \mu_i} = 176.78 \right. \text{ (Throttle control)}$$

All inputs and outputs are normalised in b/w 0 to 255

If membership fns are not symmetrical, method of height (X)
 " " COI (✓)



$$x_1 = 159 \quad A_1 = 0.125$$

$$x_2 = 191 \quad A_2 = 0.1563$$

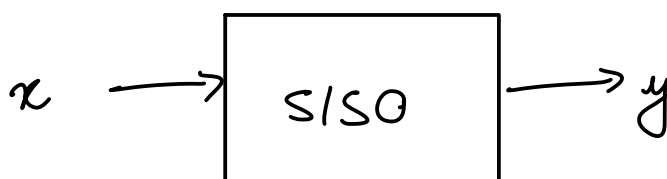
$$x^* = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = x^* = 176.6$$

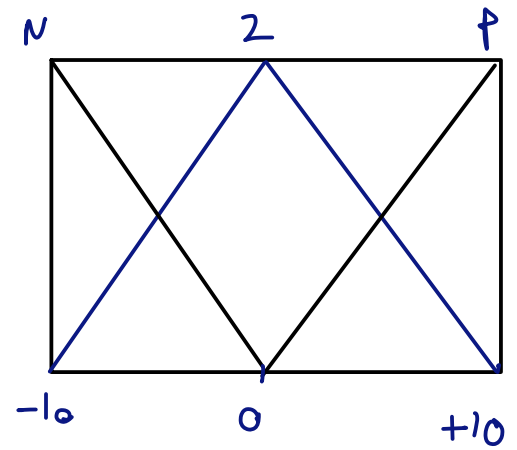
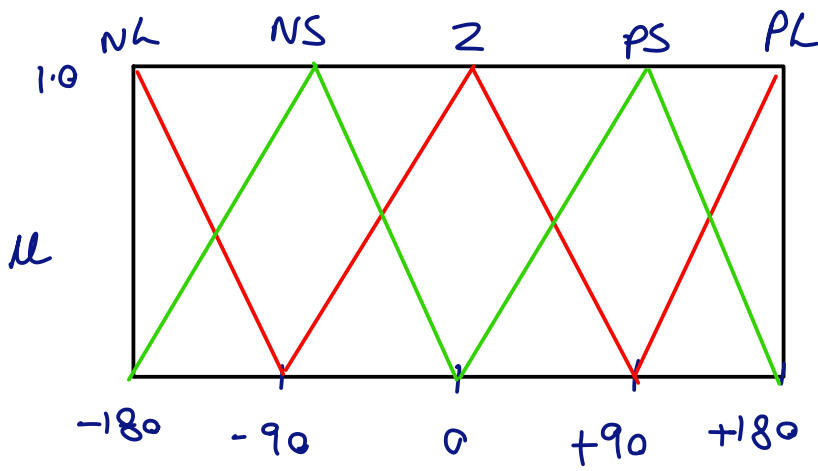
→ In symmetric cases → method of height & method of COI will give same results

MIN → applies to rule part

MAX → applies during defuzzification (for area chopping)

↳ when fuzzy output is same, choose higher value rule for area chopping





$y = \sin x \rightarrow$ actual system

$\hookrightarrow$ Not symmetric

Rule book

x	Nh	NS	Z	PS	PL
y	Z	N	Z	P	Z

$x = -150$	$y = -3.33$	Actual answer
		-5
+60	= +3.33	8.6
-50	= -5.6	-7.66
+100	= +8.9	+9.84

There's a difference b/w actual & fuzzy answers, so how to make them close $\rightarrow$ By tuning

$\hookrightarrow$ You can't change rulebook



to change membership fn. (μ values)
Classes will remain same

12/02/24 Fuzzy Logic

$\rightarrow$ Rulebook (Intelligence)

$\rightarrow$ Membership function (tuning parameter, can be changed to a different forms)

(*) if M.F is non-uniform
the sum may not be 1.



main tasks
during Fuzzification
Algorithm.

Mathematical model

1.) PDE / Equations

2.) B.C / I.C.

3.) K, ρ, C_p (may vary with phase change)

4.) Assumptions

↳ continuous casting

Take a lot of time (not employed in Real Time Systems)

∴ we use AI systems (takes time to train, but once trained, decision processing takes less time).

Gradient-free optimisation (AI types)

↳ GA (Genetic Algorithm).

↳ PSO

↳ ACO (Ant colony optimisation)

↳ SA (simulated annealing)

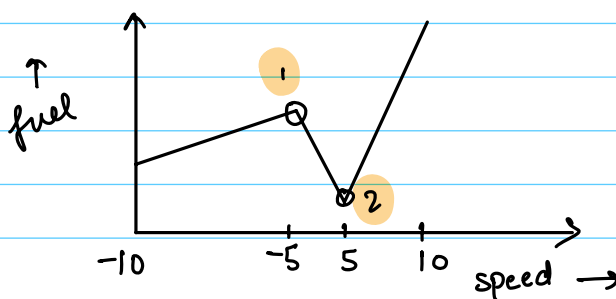
Fuzzy Rule set

Non-fuzzy rule set example:-

if $-10 \leq x \leq -5$ then $y = 0.1x + 7.0$ — ①

if $-5 \leq x \leq +5$ then $y = -0.5x + 4$ — ②

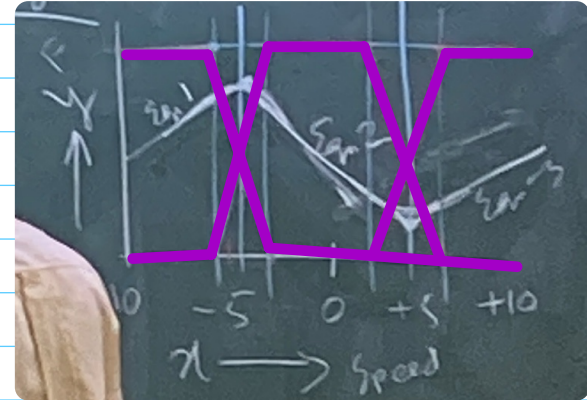
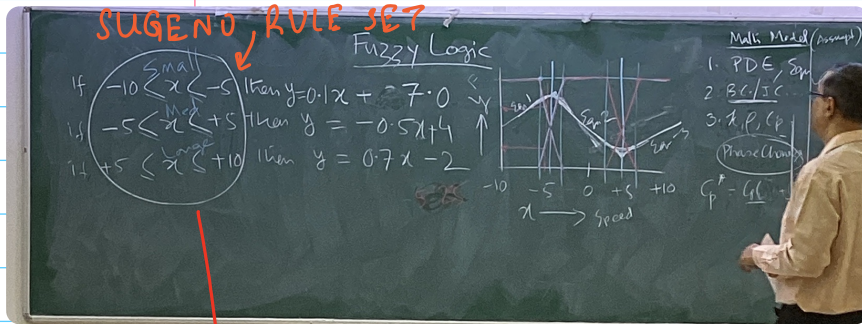
if $+5 \leq x \leq 10$ then $y = 0.7x - 2$ — ③



[Multi-point fuel injection] systems
(These curves are generated from OFFLINE simulations)

of Thermodynamic, Torque generation etc. data)

At point 1 & 2 the car engine gets jerk due to sudden change in fuel flow rate. (SMOOTHING is needed)



Fuzzy rule set
(smoothing of sharp corner)

membership function values will be weights for eq'n ①, ②, ③

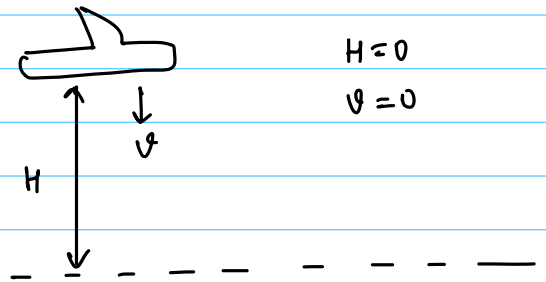
If $\sum u_i = 1$ then result is $\sum u_i x_i$

$\sum u_i \neq 1$ then $\frac{\sum u_i x_i}{\sum u_i}$

Our original rule set method (fuzzy $\rightarrow$ fuzzy)
if x is (small) then y is (small) .

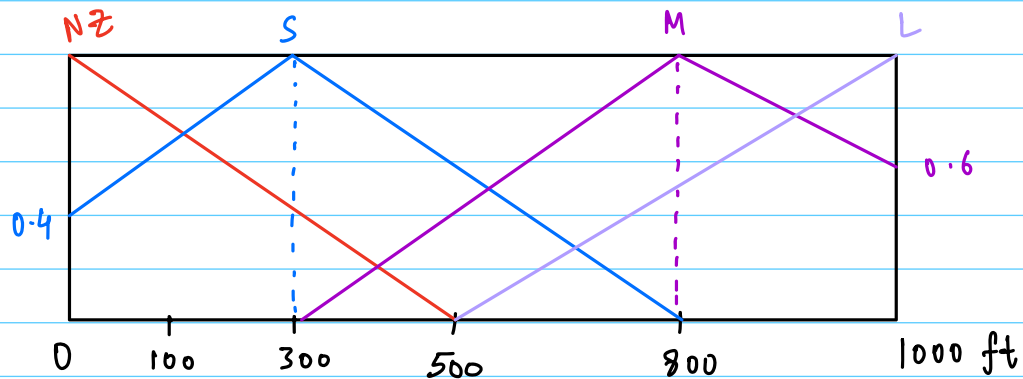
↓
"Mamdani"

13/2/24



$v \rightarrow$ FLC $\rightarrow$ control force (to control speed in y-direction)
 $H \rightarrow$

M.F
(Height)



Nz - near zero

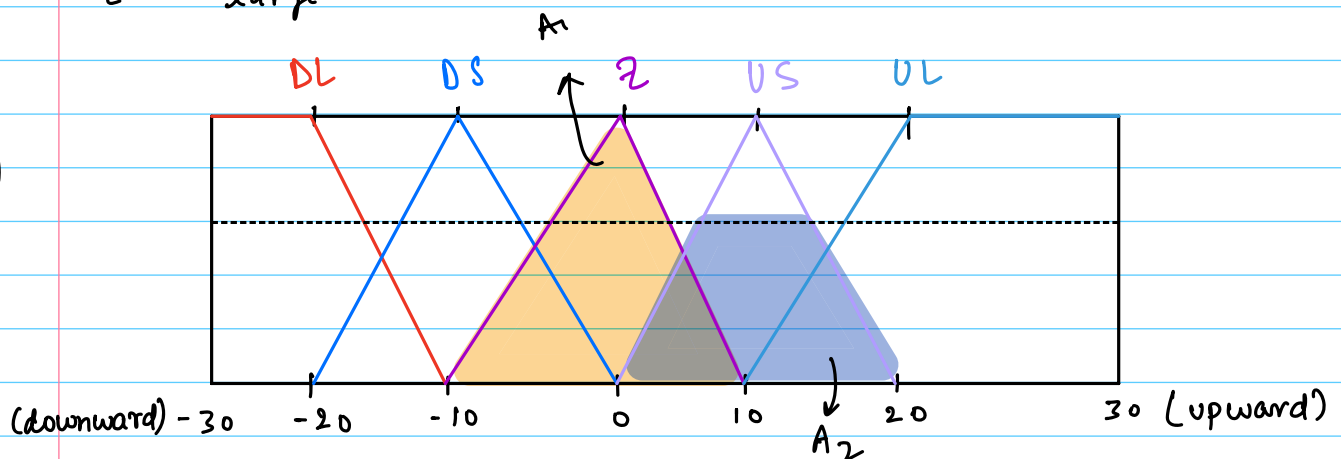
S - small

M - medium

L - large

$\left[\sum u_i \neq 1 \text{ (asymmetrical m.f)} \right]$
 (may)

M.F
(v, F)



$v \rightarrow$ ft/s

$F \rightarrow$ force limits

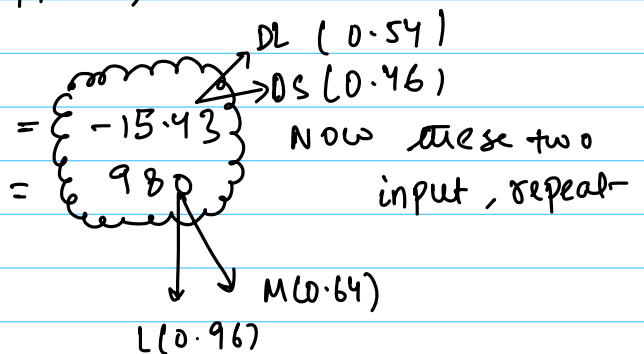
Method of height

$$(C.O) = \frac{(1 \times 0) + (0.6 \times 10)}{(1 + 0.6)} = 3.75 \text{ (2*)}$$

(#)

$$v_1 = -20 + 4.57 = -15.43$$

$$h_1 = 1000 - 20 = 980$$



Input

Output

min	L, DL	→	Z (0.54)	} →	DS 0.46 US 0.54 Z 0.54 max
min	L, DS	→	DS (0.46)		
min	M, DL	→	US (0.54)		
min	M, DS	→	Z (0.46)		

$$M/H \rightarrow 0.52$$

$$M/LG \rightarrow 0.35$$

$$v_2 = -15.43 + 0.35 = -15.08$$

$$h_2 = 980 - 15.43 = 964.57$$

Rule Base

$\begin{matrix} v, F \\ h \end{matrix}$	DL	DS	Z	US	UL
L	Z	DS	DL	DL	DL
M	US	Z	DS	DL	DL
S	UL	US	Z	DS	DL
Z	UL	UL	Z	DS	DS

If (H=M) AND (V=Z) THEN (F=DS)

$$F = ma$$

$$f = \frac{F}{m} = \frac{\Delta v}{\Delta t} \Rightarrow \boxed{f = \Delta v}$$

$$\text{If } \Delta t = 1 \text{ sec}$$

$$\Rightarrow v_{i+1} = v_i + f_i$$

$$\Rightarrow h_{i+1} = h_i + v_i$$

$$\text{Ex: } h_0 = 1000 \text{ ft}$$

$$v_0 = -20 \text{ ft/s}$$

$$F_0 = ? \text{ (from FLC)}$$

$$\begin{aligned} \text{(\#)} \quad h_0 &= 1000 \text{ ft} \begin{cases} \rightarrow L(1) \\ \rightarrow M(0.6) \end{cases} \\ v_0 &= -20 \text{ ft/s} \rightarrow DL(1) \end{aligned}$$

$$\Rightarrow F \begin{cases} \rightarrow Z(1) \\ \rightarrow US(0.6) \end{cases}$$

2 rules triggered as highlighted

$$A_1 = \frac{1}{2} \times 1 \times 20 = 10$$

$$A_2 = \frac{1}{2} \times 1 \times 20 - \frac{1}{2} \times 0.4 \times 8 = 8.4$$

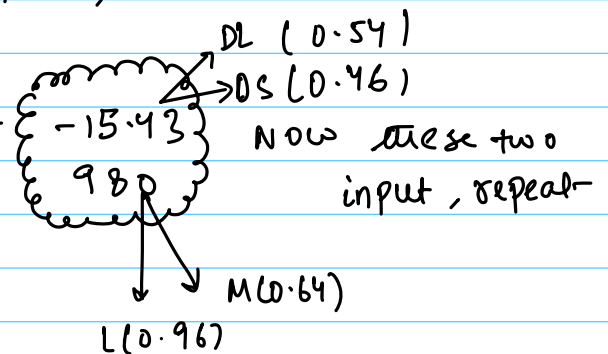
$$\text{Crisp Output} = \frac{\sum A_i x_i}{\sum A_i} = \frac{10(0) + 8.4(10)}{10 + 8.4}$$

$$= \underline{\underline{4.5652 (x^*)}}$$

Method of height

$$(C.O) = \frac{(1 \times 0) + (0.6 \times 10)}{(1 + 0.6)} = 3.75 \text{ (2*)}$$

(#) $v_1 = -20 + 4.57 = -15.43$
 $h_1 = 1000 - 20 = 980$



	Input	Output
min	L, DL →	Z (0.54)
min	L, DS →	DS (0.46)
min	M, DL →	VS (0.54)
min	M, DS →	Z (0.46)

→ DS 0.46
VS 0.54
Z 0.54 max

$$M/H \rightarrow 0.52$$

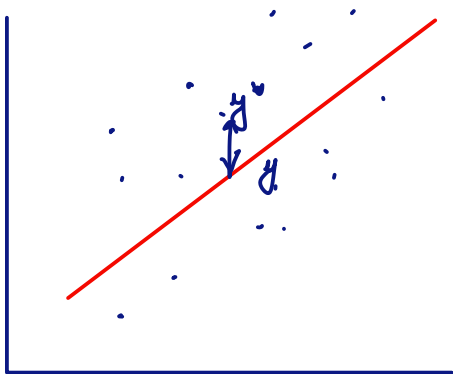
$$M/LG \rightarrow 0.35$$

$$v_2 = -15.43 + 0.35 = -15.08$$

$$h_2 = 980 - 15.43 = 964.57$$

27 Feb Tue Class odd notes

Randomise data (if we segregate based on sequence, prediction must be upto mark)
 ↳ Then do train test split



$$\sum (y^* - y)^2 \rightarrow \text{error need to minimize}$$

$$\sum (y - a - bx)^2 = S$$

$$\frac{\partial \mathcal{L}}{\partial a} = 0 \quad \frac{\partial \mathcal{L}}{\partial b} = 0$$

I/P	O/P
x	y

$w, v \rightarrow$ weights $\rightarrow$ of neural nets

$E = f(W, V) \rightarrow$ ~~over~~ fn. of weights

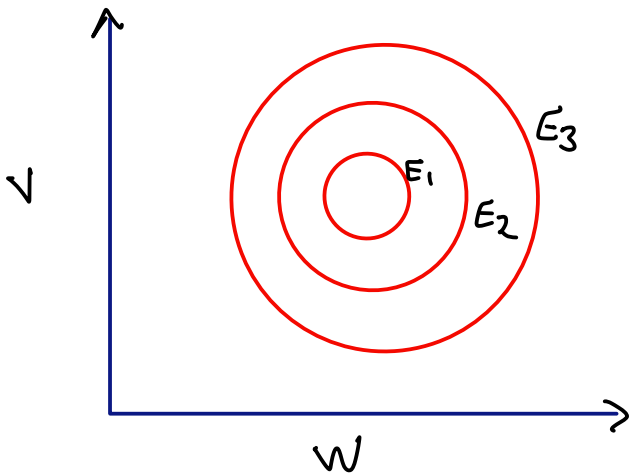
$$\frac{\partial E}{\partial \omega} = 0 \quad \frac{\partial E}{\partial v} = 0$$

$$E = RMSE$$

Handwritten diagram illustrating a machine learning workflow:

- The dataset is split into **Training** (2/3) and **Testing** (1/3).
- The **Training** set is used to train a model, represented by **X**.
- The **Testing** set is used to evaluate the model's performance.
- The model **X** takes **actual** values (y) as input and produces **predicted** values ($\hat{y}$).

$$\frac{\sum (Y - \hat{Y})^2}{N} \rightarrow \text{need to minimise}$$



$E_3 > E_2 > E_1 \rightarrow$ contours of error

Suppose we get E_3 , when we chose w, v as random, then after

Iterations we get E_2 and finally E_1
so we say we have converged
where E_1 is the tolerance and V, W

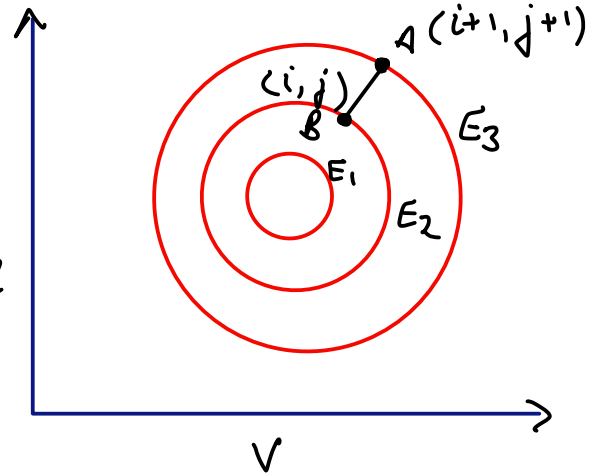
can be employed in real time

gradient descent method $\rightarrow$ at every point you have to find the steepest gradient

$$\vec{G} = \left(\frac{\partial E}{\partial v}\right) \hat{i} + \left(\frac{\partial E}{\partial w}\right) \hat{j}$$

$$\vec{AB} = -\eta [\vec{G}] \quad \eta \rightarrow \text{magnitude}$$

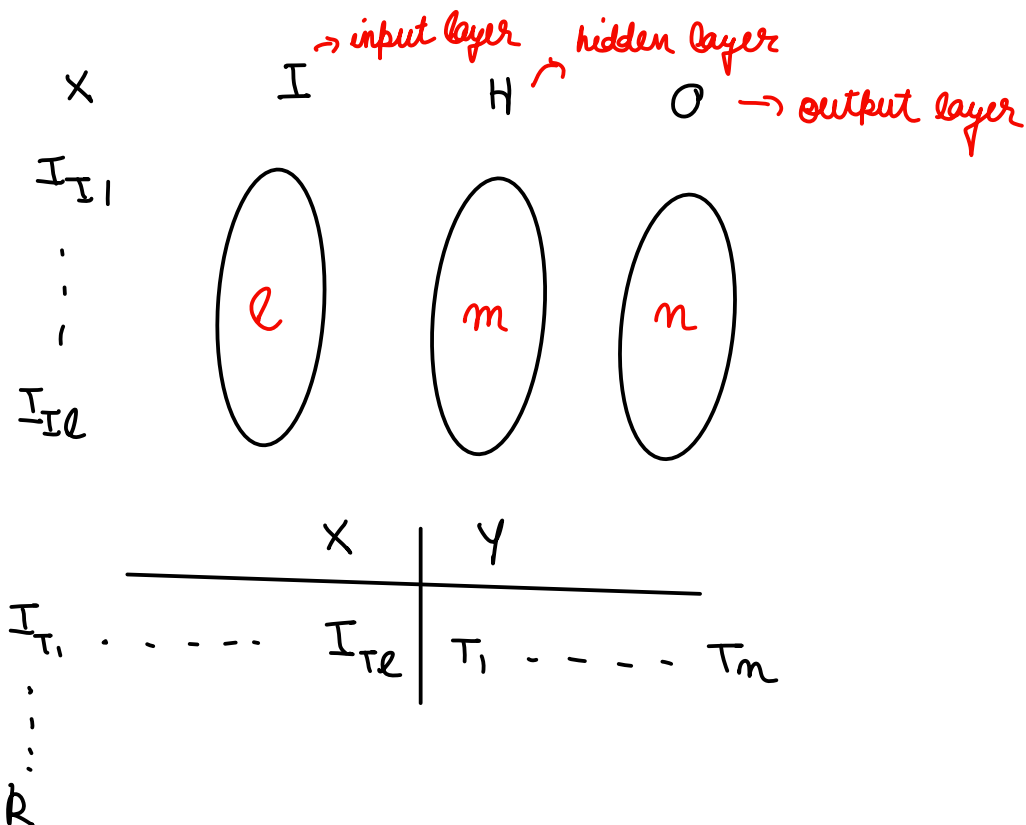
$$\begin{aligned} \vec{AB} &= (v_{i+1} - v_i) \hat{i} + (w_{j+1} - w_j) \hat{j} \\ &= (\Delta v) \hat{i} + (\Delta w) \hat{j} \end{aligned}$$



$$\begin{aligned} \Delta v &= -\eta \left(\frac{\partial E}{\partial v}\right) \\ \Delta w &= -\eta \left(\frac{\partial E}{\partial w}\right) \end{aligned}$$

 $\rightarrow \text{Most important}$

$$E_k = \frac{1}{2} [T_k - O_{ok}]^2$$



$$E_k = \frac{1}{2} [T_k - O_{ok}]^2$$

actual → predicted

$$\frac{\partial E_k}{\partial W_{ik}} = \left(\frac{\partial E_k}{\partial O_{ok}} \right) \left(\frac{\partial O_{ok}}{\partial I_{ok}} \right) \left(\frac{\partial I_{ok}}{\partial W_{ik}} \right)$$

→ why chain rule?

↓
No direct relationship b/w E & W , and we have relationship b/w E, O, O, I & I, W

$$\downarrow$$

$$-(T_k - O_{ok})$$

$$O_{ok} = \frac{1}{1 + e^{-\lambda(I_{ok})}}$$

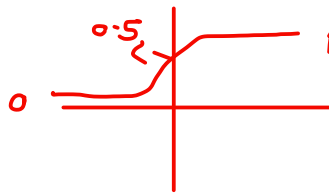
→ sigmoidal transfer fn.

↓
can be used to make training more effective

$$y = \frac{1}{1 + e^{-\lambda x}}$$

$$y \rightarrow [0, 1]$$

pass through y axis at 0.5



→ slope depends on λ

$$\frac{\partial O_{ok}}{\partial I_{ok}} = (\lambda O_{ok} (1 - O_{ok}))$$

can be changed by λ
 sensitively → changing x , y changes
 insensitively → " " , y does not change

$$I_{ok} = W_{1k} \cdot O_{H1} + W_{2k} \cdot O_{H2} + W_{ik} \cdot O_{Hi} \dots + W_{mk} \cdot O_{Hm}$$

$$\frac{\partial I_{ok}}{\partial W_{ik}} = O_{Hi}$$

$$\frac{\partial E_k}{\partial W_{ik}} = -(T_k - O_{ok}) (\lambda O_{ok} (1 - O_{ok})) (O_{Hi}) \rightarrow \text{final expression}$$

η → learning rate → if large then will diverge
 η, λ → tuning parameters

Add $5/3$, $12/3$ Notes $11/3 \rightarrow$ No class

Genetic Algorithm $\rightarrow$ bio-inspired / nature inspired, multiple initial guess points

$\hookrightarrow$ optimisation tool, gradient free $\rightarrow$ no need to take derivative
 $\hookrightarrow$ so good for non-differentiable fns.

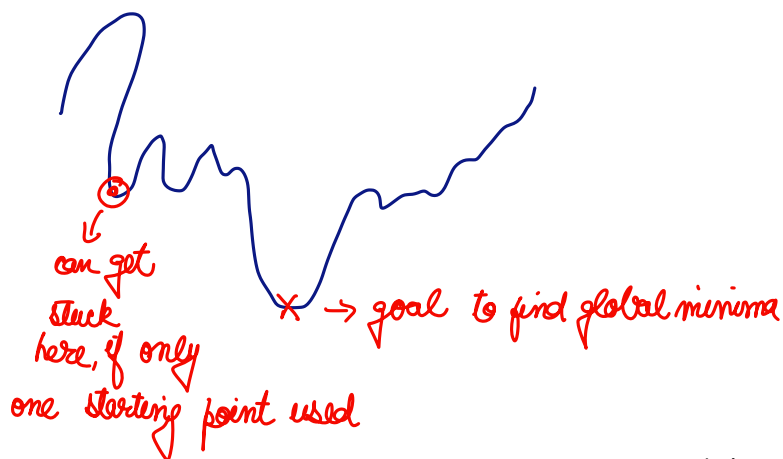
$f(x) \rightarrow$ objective fn.

subject to certain conditions $\rightarrow$ some constraints

LP $\rightarrow$ objective fn $\rightarrow$ linear and conditions are also linear

maximise $f(x)$

$\frac{\partial f}{\partial x} = 0$ subject to some conditions



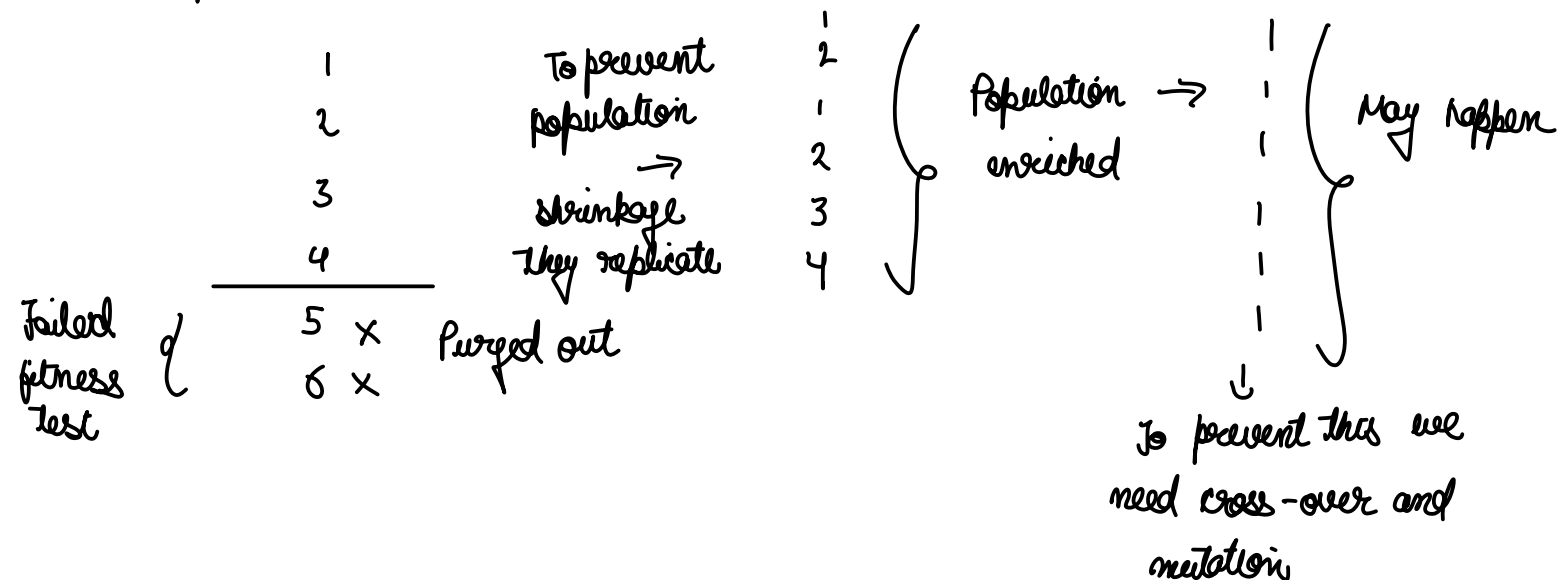
Darwin's theory

$\rightarrow$ a new species will come from already existing species by

- Reproduction
- Cross-over
- Mutation

goal: Objective function $\rightarrow$ have to optimise it

① Population



6 bit number

(A) 1 0 1 0 1 0
(B) 1 1 1 0 0 0
1 1 0 0 1 1
1 0 1 1 0 0

10 members (arbitrary)
Total $2^6 \rightarrow 64 \rightarrow$ population

want to find x
which will give max.
value of $f(x)$

If I don't do cross-over/mutation $\rightarrow$ won't be able to find max,
If I find max in 10, then it's fine, else cross-over/mutation required

Cross-over $\rightarrow$ pick 2 random numbers, randomly decide where you want to do cross-over

A1 A2
A 10 | 1 0 1 0
B B1 B2
11 | 1 0 0 0
 $\rightarrow$ Randomly select 2 numbers
 $\rightarrow$ " choose point of crossover, and do crossover
 $\hookrightarrow$ cross over place, can be anywhere

A1 B2 $\rightarrow$ 1 0 1 0 0 0
B1 A2 $\rightarrow$ 1 1 1 0 1 0

New fitness test on them

$\rightarrow$ Merit list generated $\rightarrow$

Population $\rightarrow$ undergoes fitness test $\rightarrow$ Those who fail are discarded

$\downarrow$
Toppers are replicated

(to more entries with toppers)

Mutation

$\hookrightarrow$ Randomly select 1 member

$\hookrightarrow$ Randomly select 1 bit $\rightarrow$ Flip it

ex 1 0 1 1 0 0

1 0 1 0 0 0 $\rightarrow$ Mutated, flipped the
random chosen bit
of random member

Cross-over, mutation $\rightarrow$ gives you new

members, so that you aren't stuck
with old members

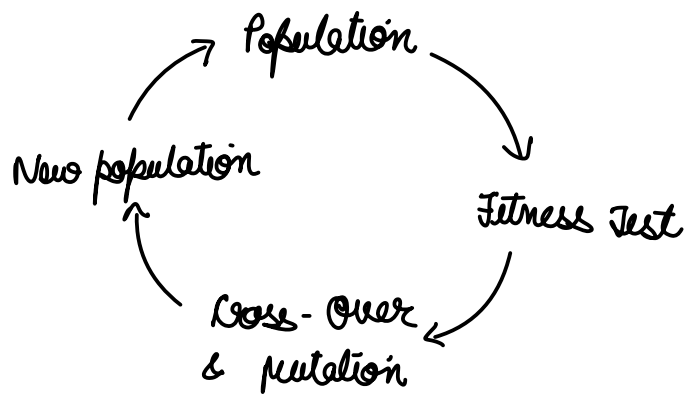
(2) Fitness Test

(3) Cross-over & Mutation

5% to 20% crossover and 1-5% mutation $\rightarrow$ Tuning Parameters

④ New population

↳ again undergo fitness test



Loop repeated
till objective fn. not
saturated
no improvement

$$\frac{C_{\min} - C_{\max}}$$

$$C_i = C_{\min} + \frac{b}{(2^L - 1)} (C_{\max} - C_{\min})$$

b = # in decimal

L = # of bit

ex $b = 18$

$L = 6$

$C_{\min} = -2$, $C_{\max} = +5$

search domain

$(-2, +5)$ → you have to search in this range

$C_i = 0$ → It's like a kind of scaling

Data-set

x	y
1	1
2	2
3	3
6	6

$$y = c_1 x + c_2$$

↳ want to fit this, so need to find c_1, c_2 so data fits the best.

Soln

12-bit string

6 bit for c_1 , 6 bit for c_2

- ① 000111010100
- ② 010010001100
- ③ 010101101010
- ④ 100100001001

Population
(Random)

so don't
know who is the
champion, who is the
loser

c_1	c_2
$7 \rightarrow -1.22$	$20 \rightarrow 0.22$
0	-0.67
0.33	2.67
2.00	-1.00

usually starting points for c_1 & c_2

mapped numbers, after scaling

If you change the range, numbers will also change

$$y_c = -1.22x + 0.22 \rightarrow \text{fitness test}$$

Put x , get calculated value of y , we have actual value of y , calculate error

$$\sum (y_c - y_t)^2 \rightarrow \text{Want to minimise this}$$

y_c - calculated

y_t - test

converted minimisation problem
to maximisation problem
(easier to converge)

$$[400 - \sum (y_c - y_t)^2] = f(x)$$

a large
number can be 500, 1000 etc

$f(x)$

purged out the bottom one

- ① $\rightarrow 147 \rightarrow$ discarded, couldn't pass the fitness test
- ② $\rightarrow 332$
- ③ $\rightarrow 391 \rightarrow$ best
- ④ $\rightarrow 358$

population shrank

so fitter replication

population with better
fitness score

will keep
on repeating
till fitness score
stagnates

If my avg.
fitness $\uparrow$

11/4/24

Post Break class

example

$$f(x) = (x_1 - 1.5)^2 + (x_2 - 4)^2$$

$$\text{subject to: } 4.5x_1 + x_2^2 - 18 \leq 0$$

$$2x_1 - x_2 - 1 \geq 0$$

$$0 \leq x_1, x_2 \leq 1$$

Calculate individual c_i

$$C = \sum c_i$$

Cross over probability = 80%.

Mutation = 10%.

$$\phi(x) =$$

$$\phi(x) = f(x) [1 + KC]$$

$$K=10$$

If $c_i = 0$ (all), then $C = 0$ $\phi(x) = f(x) \rightarrow$ all satisfied

TABLE 4.9a
1st iteration using a genetic algorithm, Example 4.6

String number	String	C_1 (bin.)	C_1	C_2 (bin.)	C_2	y_1	y_2	y_3	y_4	$f(x) = 400 - \sum (y_i - y_i)^2$	Expected count = Average	Actual count
1	000111 010100	7	-1.22	20	0.22	-1.00	-2.22	-4.67	-7.11	147.49	0.48	0
2	010010 001100	18	0.00	12	-0.67	-0.67	-0.67	-0.67	-0.67	332.22	1.08	1
3	010101 101010	21	0.33	42	2.67	3.00	3.33	4.00	4.67	391.44	1.27	2
4	100100 001000	36	2.00	9	-1.00	1.00	3.00	7.00	11.00	358.00	1.17	1
Sum										1229.15		
Average										307.29		
Maximum										391.44		

Handwritten notes on table:
 x_1 (first 6 bits)
 x_2 (last 6 bits)
 Error, min problem converted to max problem
 Mapping $C_{max} - C_{min}$ in 0-63 range
 Scaled value of C_1
 Scaled value of C_2 61/9 -2 to 5
 Squared Error

 $L=6$ (6 bits)Mapping $C_{max} - C_{min}$ in 0-63 range

$$C_i = C_{min} + \frac{b}{2^L - 1} (C_{max} - C_{min})$$

In this problem range was -2 to 5

search domain we are looking for c_1, c_2 in this

Imp If we increase the number of bits, our least count will increase, $\frac{1}{2^L - 1}$ factor will become smaller and finer hence our accuracy will increase

$$y_i = C_1 x_{i1} + C_2 x_{i2}$$

calculated Column 4, 6

Column 11 $\rightarrow$ nothing but fitness value2, 3, 4 string $\rightarrow$ above average1 " $\rightarrow$ below average $\rightarrow$ needs to be eliminated from population

Topper will be

multiplied $\rightarrow 3$

Then will go cross-over & mutation

TABLE 4.9a
1st iteration using a genetic algorithm, Example 4.6

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
String number	String	C_1 (bin.)	C_1	C_2 (bin.)	C_2	y_1	y_2	y_3	y_4	$f(x) = 400 - \sum (y_i - y_i)^2$	Expected count = $f(x)/\text{Average}$	Actual count
1	000111 010100	7	-1.22	20	0.22	-1.00	-2.22	-4.67	-7.11	147.49	0.48	0
2	010010 001100	18	0.00	12	-0.67	-0.67	-0.67	-0.67	-0.67	332.22	1.08	1
3	010101 101010	21	0.33	42	2.67	3.00	3.33	4.00	4.67	391.44	1.27	2
4	100100 001001	36	2.00	9	-1.00	1.00	3.00	7.00	11.00	358.00	1.17	1
Sum										1229.15		
Average										307.29		
Maximum										391.44		

TABLE 4.9b
Second iteration using a genetic algorithm, Example 4.6

Selected strings	New strings	C_1 (bin.)	C_1	C_2 (bin.)	C_2	y_1	y_2	y_3	y_4	$f(x) = 400 - \sum (y_i - y_i)^2$	Expected count = $f(x)/\text{Average}$	Actual count
010101 101010	010110 001100	22	0.44	12	-0.67	-0.22	0.22	1.11	2.00	375.78	1.15	1
010010 001100	010001 101010	17	-0.11	42	2.67	2.56	2.44	2.22	2.00	380.78	1.17	2
010101 101010	010101 101001	21	0.33	41	2.56	2.89	3.22	3.89	4.56	292.06	0.90	1
100100 001001	100100 001010	36	2.0	10	-0.89	1.11	3.11	7.11	11.11	255.73	0.78	0
Sum										1304.35		
Average										326.09		
Maximum										380.78		

3

For above example, $f(x), \phi(x)$

Plug in x_1, x_2 in $g_i(x)$ equations $\rightarrow C_i = 0$ or $g_i(x)$

$$C = \sum C_i$$

$\hookrightarrow$ use this to calculate $\phi(x)$

$$\hookrightarrow f(x)[1 + K \cdot C]$$

2/4, 8/4 add